

Chapter 9: The Behavior of Solutions Part II

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Outlines

9.6 The Properties of **Ideal** Solutions

- 9.6.1 The Change in Volume Accompanying the Formation of an Ideal Solution
- 9.6.2 The Enthalpy of Formation of an Ideal Solution
- 9.6.3 The Entropy of Formation of an Ideal Solution

9.7 **Non-Ideal** Solutions

9.8 Application of the Gibbs–Duhem Relation to the Determination of **Activity**

- 9.8.1 The Relationship between Henry's and Raoult's Laws
- 9.8.2 Direct Calculation of the Total Molar Gibbs Free Energy of Mixing

9.6. The Properties of Ideal Solutions

- ✓ The components of a Raoultian solution obey the relation $a_i = X_i$. An **ideal** solution is a solution which obeys Raoult's law at all temperatures, pressures, and compositions.
- ✓ For an **ideal** binary A–B solution, the Gibbs free energy of mixing is

$$\begin{array}{c} \Delta G^M = RT (X_A \ln a_A + X_B \ln a_B) \\ \downarrow \qquad \qquad \qquad \downarrow \\ \Delta G^{M,id} = RT (X_A \ln X_A + X_B \ln X_B) \end{array}$$

The **partial** Gibbs free energy of mixing of the components A and B given as

$$\overline{\Delta G}_A^{M,id} = RT \ln X_A \qquad \overline{\Delta G}_B^{M,id} = RT \ln X_B$$

9.6.1. The **Change in Volume** Accompanying the Formation of an Ideal Solution

- ✓ The general thermodynamic relationships between the state properties of a system are applicable to the partial molar properties of the components of a system.
- ✓ For the species *i* occurring in a solution,

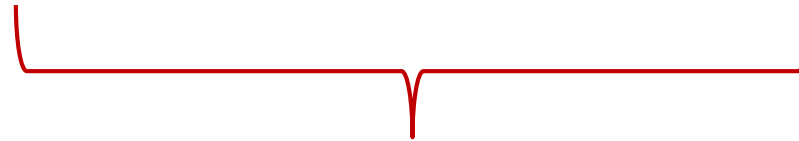
$$\left[\frac{\partial \bar{G}_i}{\partial P} \right]_{T, comp} = \bar{V}_i$$

- ✓ For pure *i*

$$\left[\frac{\partial G_i^\circ}{\partial P} \right]_{T, comp} = V_i^0$$

9.6.1. The **Change in Volume** Accompanying the Formation of an Ideal Solution

$$\left[\frac{\partial \bar{G}_i}{\partial P} \right]_{T,comp} = \bar{V}_i \qquad \left[\frac{\partial G_i^\circ}{\partial P} \right]_{T,comp} = V_i^0$$



$$\left[\frac{\partial (\bar{G}_i - G_i^\circ)}{\partial P} \right]_{T,comp} = (\bar{V}_i - V_i^0)$$



$$\left[\frac{\partial \Delta \bar{G}_i^M}{\partial P} \right]_{T,comp} = \Delta \bar{V}_i^M$$

9.6.1. The **Change in Volume** Accompanying the Formation of an Ideal Solution

$$\left[\frac{\partial \Delta \bar{G}_i^M}{\partial P} \right]_{T, comp} = \Delta \bar{V}_i^M$$

- ✓ The **change in volume** due to mixing, ΔV^M , is the difference between the volumes of the **components in the solution** and the volumes of the **pure components**; that is,
- ✓ For a binary A – B solution containing n_A moles of A and n_B moles of B ,

$$\begin{aligned} \Delta V'^M &= \left(n_A \bar{V}_A + n_B \bar{V}_B \right) - \left(n_A V_A^\circ + n_B V_B^\circ \right) \\ &= n_A \left(\bar{V}_A - V_A^\circ \right) + n_B \left(\bar{V}_B - V_B^\circ \right) \\ &= n_A \Delta \bar{V}_A^M + n_B \Delta \bar{V}_B^M \end{aligned}$$

9.6.1. The **Change in Volume** Accompanying the Formation of an Ideal Solution

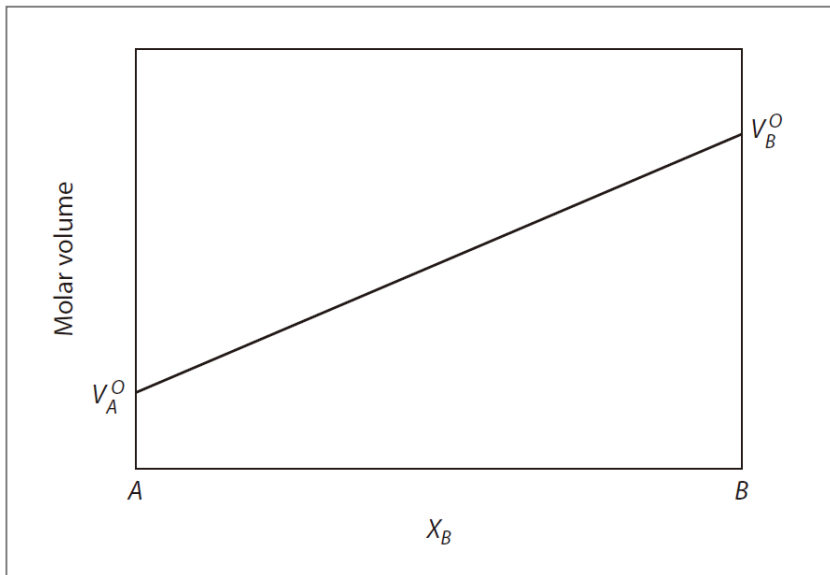
- ✓ In an ideal solution, $\Delta \bar{G}_i^{M, id} = RT \ln X_i$, and since this term is **not** a function of pressure, $\Delta \bar{V}_i^{M, id} = 0$.
- ✓ Thus, it is seen that the change in volume accompanying an ideal solution is zero; that is,

$$\Delta V'^{M, id} = 0$$

- ✓ The **volume** of an **ideal solution** is thus equal to the **sum of** the volumes of the **pure components**.

9.6.1. The **Change in Volume** Accompanying the Formation of an Ideal Solution

- ✓ At any composition, the values of the **partial molar volumes** $\bar{V}_A^M$ and $\bar{V}_B^M$ are obtained as the **intercepts of the tangents** to the volume–composition line with the respective axes. Since the molar volume of an ideal solution is a **linear function of composition**, the tangent at any point coincides with the straight line, such that



$$\bar{V}_A = V_A^0 \quad \bar{V}_B = V_B^0$$

$$\Delta V'^{M,id} = 0$$

Figure 9.6
the variation, with composition, of the molar volume of a binary ideal solution.

9.6.2. The **Enthalpy** (H) of Formation of an Ideal Solution

- ✓ For a component in a solution, the Gibbs–Helmholtz equation, can be used to obtain

$$\left[\partial \left(\frac{G}{T} \right) / \partial T \right]_P = -\frac{H}{T^2} \xrightarrow{\text{@ const. } P, \text{ comp}} \left[\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right]_{P, \text{ comp}} = -\frac{\bar{H}_i}{T^2}$$

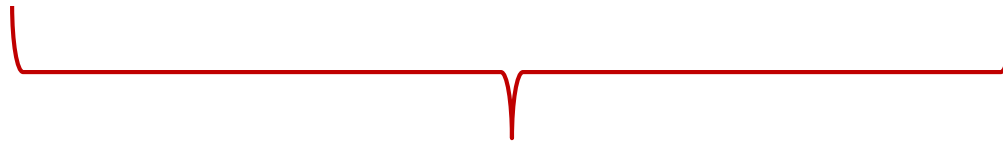
- ✓ For the **pure** component,

$$\left[\partial \left(\frac{G}{T} \right) / \partial T \right]_P = -\frac{H}{T^2} \xrightarrow{\text{@ const. } P, \text{ comp}} \left[\partial \left(\frac{G_i^\circ}{T} \right) / \partial T \right]_{P, \text{ comp}} = -\frac{H_i^\circ}{T^2}$$

$\bar{H}_i$ and H_i° are the partial molar enthalpy of i in the solution and the standard molar enthalpy of i .

9.6.2. The **Enthalpy** (H) of Formation of an Ideal Solution

$$\left[\partial \left(\frac{\bar{G}_i}{T} \right) / \partial T \right]_{P, \text{comp}} = -\frac{\bar{H}_i}{T^2} \quad \left[\partial \left(\frac{G_i^\circ}{T} \right) / \partial T \right]_{P, \text{comp}} = -\frac{H_i^\circ}{T^2}$$



$$\left[\partial \left(\frac{\bar{G}_i - G_i^\circ}{T} \right) / \partial T \right]_{P, \text{comp}} = -\frac{(\bar{H}_i - H_i^\circ)}{T^2}$$



$$\left[\partial \left(\frac{\Delta \bar{G}_i^M}{T} \right) / \partial T \right]_{P, \text{comp}} = -\frac{\Delta \bar{H}_i^M}{T^2}$$

$\Delta \bar{H}_i^M$ is the partial molar enthalpy of mixing of solution of i .

9.6.2. The **Enthalpy** (H) of Formation of an Ideal Solution

$$\left[\partial \left(\frac{\Delta \bar{G}_i^M}{T} \right) / \partial T \right]_{P, comp} = - \frac{\Delta \bar{H}_i^M}{T^2}$$

In an ideal solution,  $\Delta \bar{G}_i^{M, id} = RT \ln X_i$

$$\frac{d(R \ln X_i)}{dT} = - \frac{\Delta \bar{H}_i^M}{T^2}$$

- ✓ Since the first term is **not a function of temperature**, it is seen that, for a component of an ideal solution,

$$\Delta \bar{H}_i^{M, id} = \bar{H}_i - H_i^\circ = 0 \quad \text{or} \quad \bar{H}_i = H_i^\circ$$

9.6.2. The **Enthalpy** (H) of Formation of an Ideal Solution

- ✓ The enthalpy of formation of a solution (or the enthalpy of mixing of the components) is the difference between the enthalpies of the components in solution and the enthalpies of the pure components before mixing. Thus, for a mixture of n_A moles of A and n_B moles of B ,

$$\begin{aligned}\Delta H'^M &= \left(n_A \overline{H}_A + n_B \overline{H}_B \right) - \left(n_A H_A^\circ + n_B H_B^\circ \right) \\ &= n_A \left(\overline{H}_A - H_A^\circ \right) + n_B \left(\overline{H}_B - H_B^\circ \right) \\ &= n_A \Delta \overline{H}_A^M + n_B \Delta \overline{H}_B^M \\ \Delta H^M &= X_A \Delta \overline{H}_A^M + X_B \Delta \overline{H}_B^M\end{aligned}$$

- ✓ For an ideal solution, $\Delta \overline{H}_i^M = 0$, and thus, it is seen that the enthalpy of formation (or the enthalpy of mixing) of an ideal solution is zero; that is,

$$\Delta H^{M,id} = 0$$

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

$$dG' = -S'dT + V'dP + \sum \mu_i dn_i$$

@ const. P , comp

$$\left[\frac{\partial G}{\partial T} \right]_{P, \text{comp}} = -S \quad \left[\frac{\partial \Delta G^M}{\partial T} \right]_{P, \text{comp}} = -\Delta S^M$$

$$\Delta G^{M, id} = RT (X_A \ln X_A + X_B \ln X_B)$$

The entropy of formation of an ideal binary solution is independent of temperature and is **positive**.

$$\begin{aligned} \Delta S^{M, id} &= - \left[\frac{\partial \Delta G^{M, id}}{\partial T} \right]_{P, \text{comp}} \\ &= -R (X_A \ln X_A + X_B \ln X_B) \end{aligned}$$

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

- ✓ Total change in configurational entropy gives, for the mixing of N_A particles of A with N_B particles of B,

$$\begin{aligned}\Delta S'^M &= k_B \ln \frac{(N_A + N_B)!}{N_A! N_B!} \\ &= k_B [\ln(N_A + N_B)! - \ln N_A! - \ln N_B!]\end{aligned}$$

- ✓ Application of Stirling's theorem* (after James Sterling, 1692–1770) gives

$$\begin{aligned}\Delta S'^M &= k_B [(N_A + N_B) \ln(N_A + N_B) - (N_A + N_B) - N_A \ln N_A + N_A - N_B \ln N_B + N_B] \\ &= -k_B \left[N_A \ln \left(\frac{N_A}{N_A + N_B} \right) + N_B \ln \left(\frac{N_B}{N_A + N_B} \right) \right]\end{aligned}$$

* Stirling's theorem is $m! = \sqrt{2\pi m} m^m e^{-m}$, and thus, $\ln m! = \ln(2\pi m) + m \ln m - m$, which, for large values of m , can be written as $\ln m! = m \ln m - m$.

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

$$\Delta S'^M = -k_B \left[N_A \ln \left(\frac{N_A}{N_A + N_B} \right) + N_B \ln \left(\frac{N_B}{N_A + N_B} \right) \right]$$

$$\frac{N_A}{N_A + N_B} = \frac{n_A}{n_A + n_B} = X_A \qquad \frac{N_B}{N_A + N_B} = X_B$$

$$N_A \text{ particles of A} = \frac{N_A}{N_0} \text{ moles of A} = n_A \text{ moles of A}$$

$$N_B \text{ particles of B} = \frac{N_B}{N_0} \text{ moles of B} = n_B \text{ moles of B}$$

N_0 is Avogadro's number.

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

$$\Delta S'^M = -k_B \left[N_A \ln \left(\frac{N_A}{N_A + N_B} \right) + N_B \ln \left(\frac{N_B}{N_A + N_B} \right) \right]$$



$$\Delta S'^M = -\underline{k_B N_0} (n_A \ln X_A + n_B \ln X_B)$$

- ✓ Since Boltzmann's constant, k_B , times Avogadro's number, N_0 , equals the gas constant, R ,

$$k_B N_0 = R$$

$$\Delta S'^M = -R (n_A \ln X_A + n_B \ln X_B)$$

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

$$\Delta S'^M = -R(n_A \ln X_A + n_B \ln X_B)$$

Division by the total number of moles, $n_A + n_B$, gives

$$\Delta S^M = -R(X_A \ln X_A + X_B \ln X_B)$$

- ✓ The increase in entropy accompanying the formation of 1 mole of an ideal solution is a measure of the increase in the number of *spatial configurations* which become available to the system as a result of the mixing process.
- ✓ This is *dependent only on* the numbers of moles of the components in the solution and is independent of temperature.

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

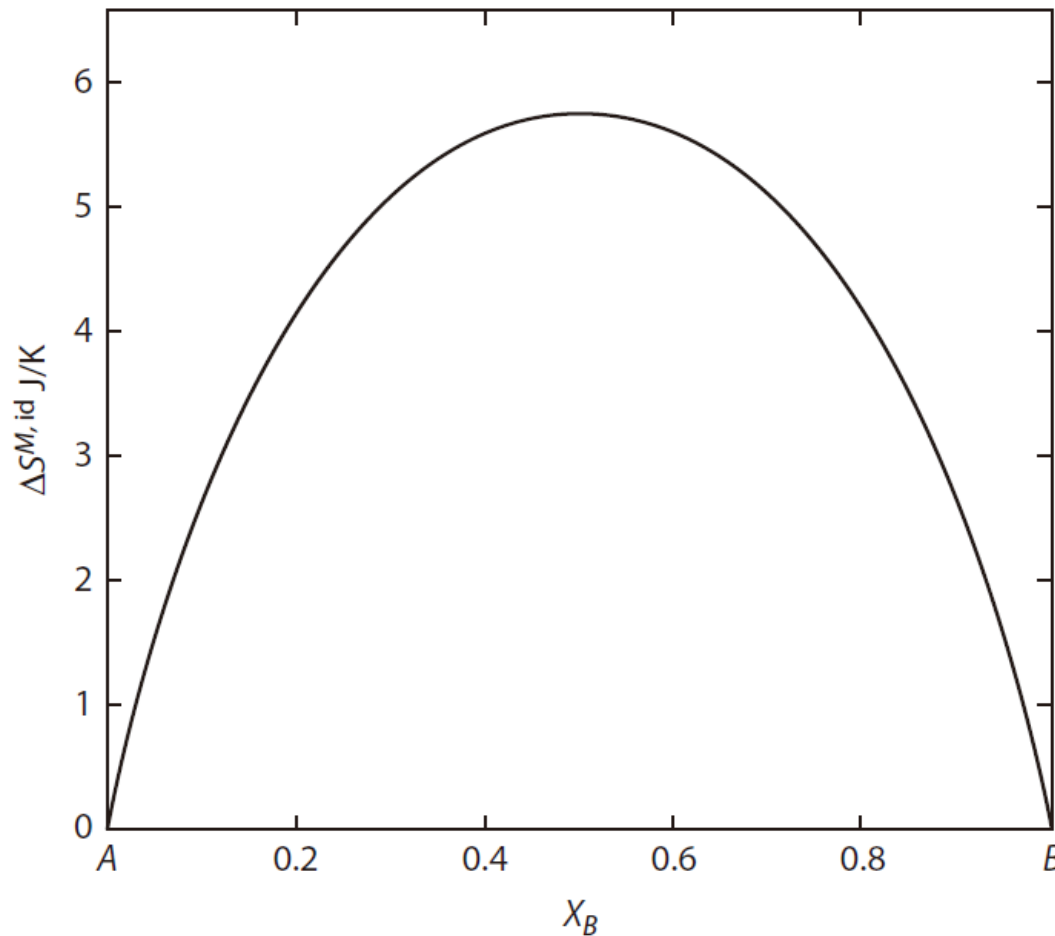


Figure 9.7 The variation, with composition, of the molar entropy of formation of a binary ideal solution. The maximum value is $R \ln 2$.

9.6.3. The **Entropy** (S) of Formation of an Ideal Solution

$$\Delta S^M = X_A \Delta \bar{S}_A^M + X_B \Delta \bar{S}_B^M$$

it is seen that, in an ideal solution,

$$\Delta \bar{S}_A^{M,id} = -R \ln X_A \quad \Delta \bar{S}_B^{M,id} = -R \ln X_B$$

For any solution,

$$\Delta G^M = \Delta H^M - T \Delta S^M$$

For an ideal solution, since $\Delta H^{M,id} = 0$, then

$$\Delta G^{M,id} = -T \Delta S^{M,id} < 0$$

$\Delta S^{M,id}$ is positive

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9.7 Non-Ideal Solutions

9.8 Application of the Gibbs–Duhem Relation to the Determination of Activity

- 9.8.1 The Relationship Between Henry's and Raoult's Laws
- 9.8.2 Direct Calculation of the Total Molar Gibbs Free Energy of Mixing

9.7. Nonideal Solutions

- ✓ A **nonideal solution** is one in which the activities of the components are not equal to their mole fractions at all compositions or temperatures.
- ✓ However, in view of the convenience of the concept of activity and the simplicity of Raoult's law, it is convenient to define an additional thermodynamic function called the **activity coefficient**, γ .
- ✓ The activity coefficient of a component of a solution is defined as **the ratio of the activity of the component to its mole fraction**; that is,
- ✓ for the component i ,

$$\gamma_i = \frac{a_i}{X_i}$$

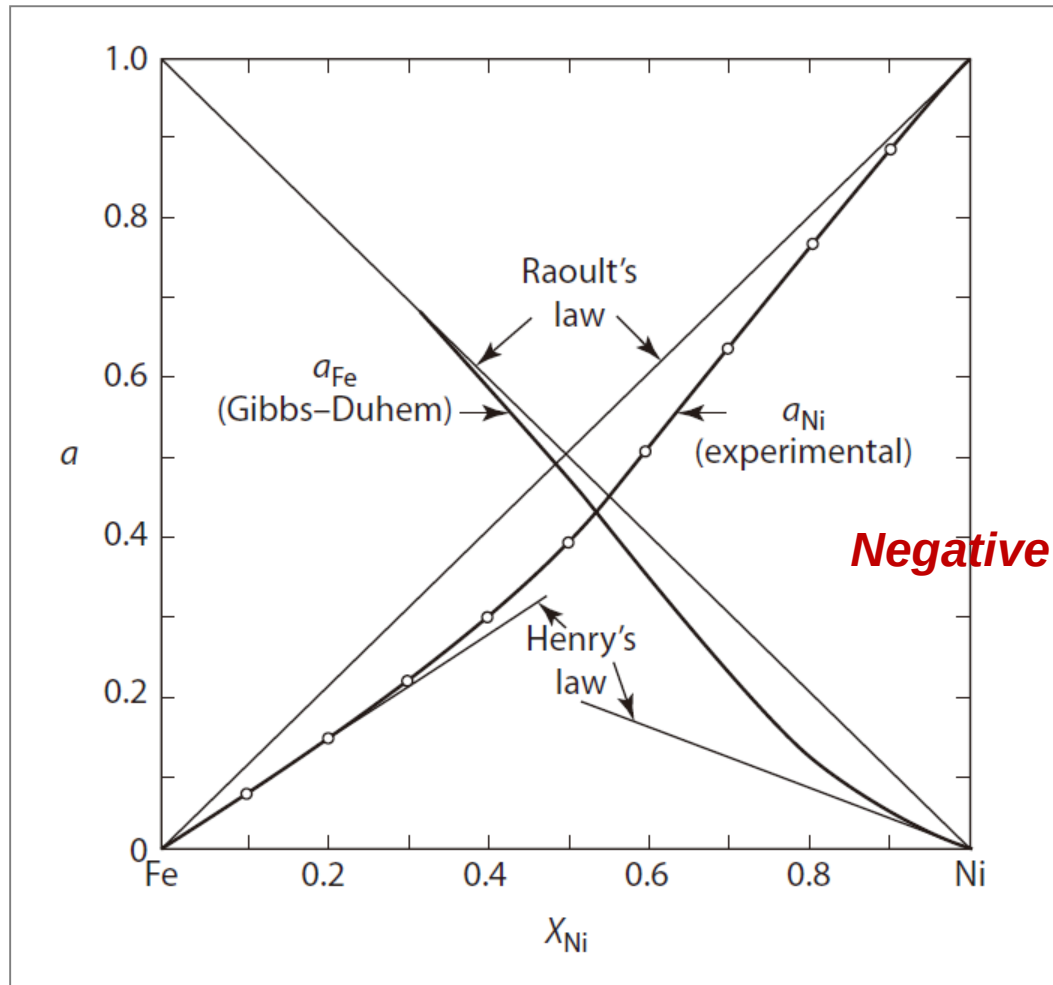
9.7. Nonideal Solutions

- ✓ The value of γ_i can be greater or less than unity ($\gamma_i = 1$ gives Raoultian behavior).

$$\gamma_i = \frac{a_i}{X_i}$$

- If $\gamma_i > 1$, then the component i is said to exhibit a **positive deviation** from Raoultian behavior,
- If $\gamma_i < 1$, then the component i is said to exhibit a **negative deviation** from Raoult's law.

9.7. Nonideal Solutions



Negative deviation: Fe-Ni

Figure 9.8 Activities in the liquid system iron–nickel at 1600° C which display negative deviation from ideality. (From G. R. Zellars, S. L. Payne, J. P. Morris, and R. L. Kipp, “The Activities of Iron and Nickel in Liquid Fe–Ni Alloys,” *Trans. AIME* (1959), vol. 215, p. 181.)

9.7. Nonideal Solutions

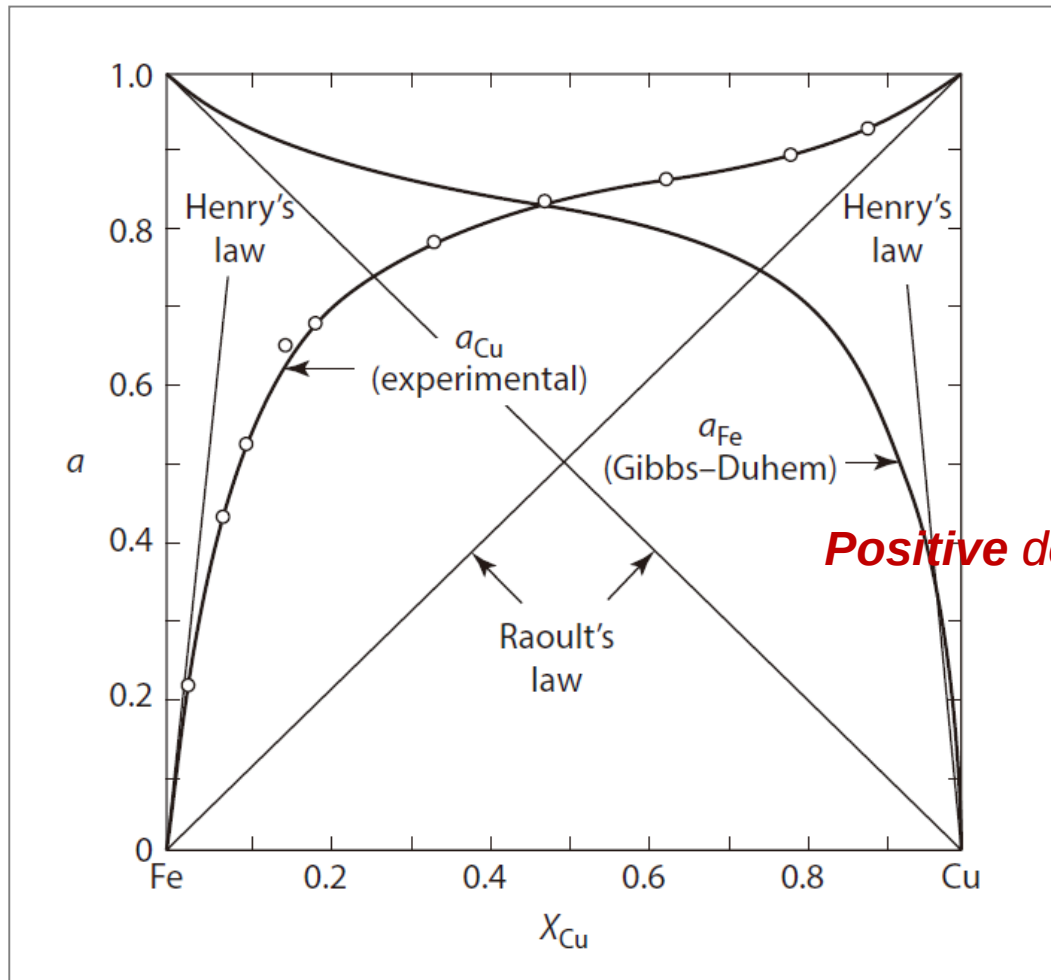


Figure 9.9 Activities in the liquid system iron– copper at 1550° C which display positive deviation from ideality. (From J. P. Morris and G. R. Zellars, “ V apor Pressure of Liquid Copper and Activities in Liquid Fe–Cu Alloys,” *Trans. AIME* (1956), vol. 206, p. 1086.)

9.7. Nonideal Solutions

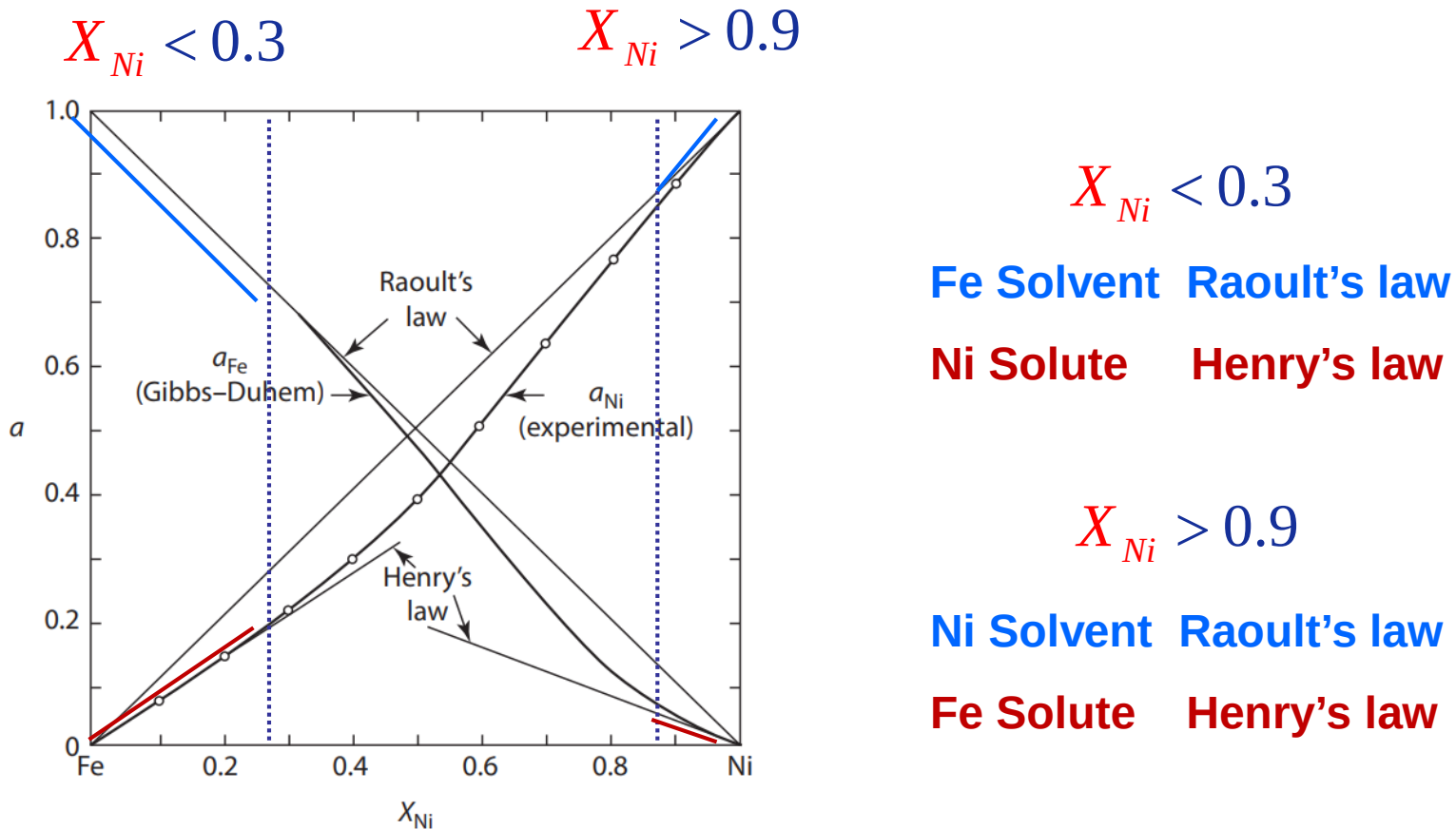


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$$\gamma_i < 1$$

Negative deviation: Fe-Ni

9.7. Nonideal Solutions

$$\gamma_i = \frac{a_i}{X_i}$$

- ✓ If γ_i varies with temperature, then $\Delta \bar{H}_i^M$ has a nonzero value; that is, from above results,

$$\frac{\partial \left(\Delta \bar{G}_i^M / T \right)}{\partial T} = - \frac{\Delta \bar{H}_i^M}{T^2}$$

$$\Delta \bar{G}_i^M = RT \ln a_i = RT \ln \gamma_i + RT \ln X_i$$

$$\frac{\partial \left(\frac{\Delta \bar{G}_i^M}{T} \right)}{\partial T} = \frac{\partial (R \ln \gamma_i)}{\partial T} = - \frac{\Delta \bar{H}_i^M}{T^2}$$



NOT varies with T

9.7. Nonideal Solutions

$$\frac{\partial(R \ln \gamma_i)}{\partial T} = -\frac{\Delta \overline{H}_i^M}{T^2}$$
$$d\left(\frac{1}{T}\right) = -\frac{dT}{T^2}$$

$$\frac{\partial(R \ln \gamma_i)}{\partial(1/T)} = \Delta \overline{H}_i^M$$

- ✓ In general, increasing the temperature of a nonideal solution causes a decrease in the extent to which its components deviate from ideal behavior; that is,
 - If $\gamma_i > 1$, then an **increase in** temperature causes γ_i to decrease **toward unity**,
 - If $\gamma_i < 1$, an **increase in** temperature causes γ_i to increase **toward unity**.

9.7. Nonideal Solutions

CASE (1) when $\gamma_i > 1$

- ✓ Thus, in a solution, the components of which exhibit **positive deviations** from ideality, the values of the activity coefficients decrease with increasing temperature.

$$\frac{\partial(R \ln \gamma_i)}{\partial\left(\frac{1}{T}\right)} = \Delta \overline{H}_i^M$$

- ✓ The **partial molar enthalpies** of formation of the components are **positive** quantities. This means that the molar enthalpy of formation (mixing) of the solution, ΔH^M , is a positive quantity, which indicates that the mixing process is **endothermic**.

9.7. Nonideal Solutions

CASE (1) when $\gamma_i > 1$

$$\frac{\partial(R \ln \gamma_i)}{\partial\left(\frac{1}{T}\right)} = \Delta \overline{H}_i^M$$

✓ ΔH^M is the quantity of thermal energy absorbed from the thermal reservoir surrounding the solution per mole of solution formed at the temperature T .

9.7. Nonideal Solutions

CASE (2) when $\gamma_i < 1$

- ✓ In a solution, the components of which exhibit **negative deviations** from ideality, the activity coefficients **increase** with increasing temperature, and hence, the **partial molar enthalpies** of mixing and the molar enthalpy of mixing are **negative**.

$$\frac{\partial(R \ln \gamma_i)}{\partial\left(\frac{1}{T}\right)} = \Delta \overline{H}_i^M$$

- ✓ Such a solution forms **exothermically**, and ΔH^M is the enthalpy **absorbed by** the constant-temperature thermal reservoir, per mole of solution formed, at the temperature T .

9.7. Nonideal Solutions

CASE (2) when $\gamma_i < 1$

$$\frac{\partial(R \ln \gamma_i)}{\partial\left(\frac{1}{T}\right)} = \Delta \overline{H}_i^M$$

- ✓ These trends can be understood by assuming that the interactions among particles decrease as the temperature rises, because the particles have greater movement with increasing temperature and therefore less chance of interacting with each other.

9.7. Nonideal Solutions

Negative deviation: Fe-Ni

- ✓ **Exothermic mixing** in an $A-B$ binary condensed system occurs when the $A-B$ bond energy is more negative than both the $A-A$ and $B-B$ bond energies, and this causes a tendency toward atomic **ordering** in the solution, in which the A atoms (B atoms) attempt to have as many B atoms (A atoms) as the nearest neighbors as is possible for the overall composition of the solution.
- ✓ Exothermic mixing thus indicates a tendency toward the formation of an atomically ordered phase or compound between the two components (e.g., the **Fe-Ni** system).

9.7. Nonideal Solutions

Positive deviation : Fe-Cu

- ✓ Conversely, **endothermic mixing** occurs when the $A-B$ bond energy is less negative than both the $A-A$ and $B-B$ bond energies, and this causes a tendency toward phase separation or **clustering** in the solution (e.g., the Fe-Cu system). The A atoms attempt to be coordinated only by A atoms, and the B atoms attempt to be coordinated only by B atoms.

9.7. Nonideal Solutions

- ✓ In both of these cases (exothermic and endothermic), the equilibrium configuration of the solution is reached as a compromise between the enthalpy and entropy factors.
- ✓ The **enthalpy**, being determined by the relative magnitudes of the bond energies, attempts to either completely order (negative deviation from ideality) or completely unmix (positive deviation from ideality) the solution. The **entropy** factor attempts to **maximize the randomness** of mixing of the atoms in the solution.

9.7. Nonideal Solutions

- ✓ For the **clustering** case, as the temperature approaches 0 K, the equilibrium state approaches that of pure A and pure B, and hence, as the temperature approaches 0 K, the configurational entropy of a solid solution approaches zero.
- ✓ For the **ordering** case, the situation is more complex and will be discussed later in Chapter 10. However, it too predicts zero configurational entropy for the crystalline alloy as the temperature approaches 0 K. These are Third Law constraints on the ground state (0 K) equilibrium entropy of alloys.

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9.8 Application of the Gibbs–Duhem Relation to the Determination of Activity

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9.8. Application of the Gibbs-Duhem Relation to the Determination of Activity

$$\sum_i X_i d\bar{Q}_i = 0 \qquad \Delta\bar{G}_i^M = \bar{G}_i - G_i^\circ = RT \ln a_i$$


$$X_A d\Delta\bar{G}_A^M + X_B d\Delta\bar{G}_B^M = 0$$

✓ Since $\Delta\bar{G}_i^M = RT \ln a_i$ then

$$X_A d \ln a_A + X_B d \ln a_B = 0$$

$$X_A d \log a_A + X_B d \log a_B = 0$$

$$d \ln a_A = -\frac{X_B}{X_A} d \ln a_B$$

$$d \log a_A = -\frac{X_B}{X_A} d \log a_B$$

9.8. Application of the Gibbs-Duhem Relation to the Determination of Activity

- ✓ If the variation of a_B with composition is known, then the integration of Equation 9.50b from $X_A = 1$ to X_A gives the value of $\log a_A$ at X_A as

$$\log a_A \text{ at } X_A = - \int_{\log a_B \text{ at } X_A=1}^{\log a_B \text{ at } X_A=X_A} \left(\frac{X_B}{X_A} \right) d \log a_B$$

- ✓ Since an analytical expression for the variation of the activity of B is not usually computed, Equation 9.51 is solved by graphical integration.

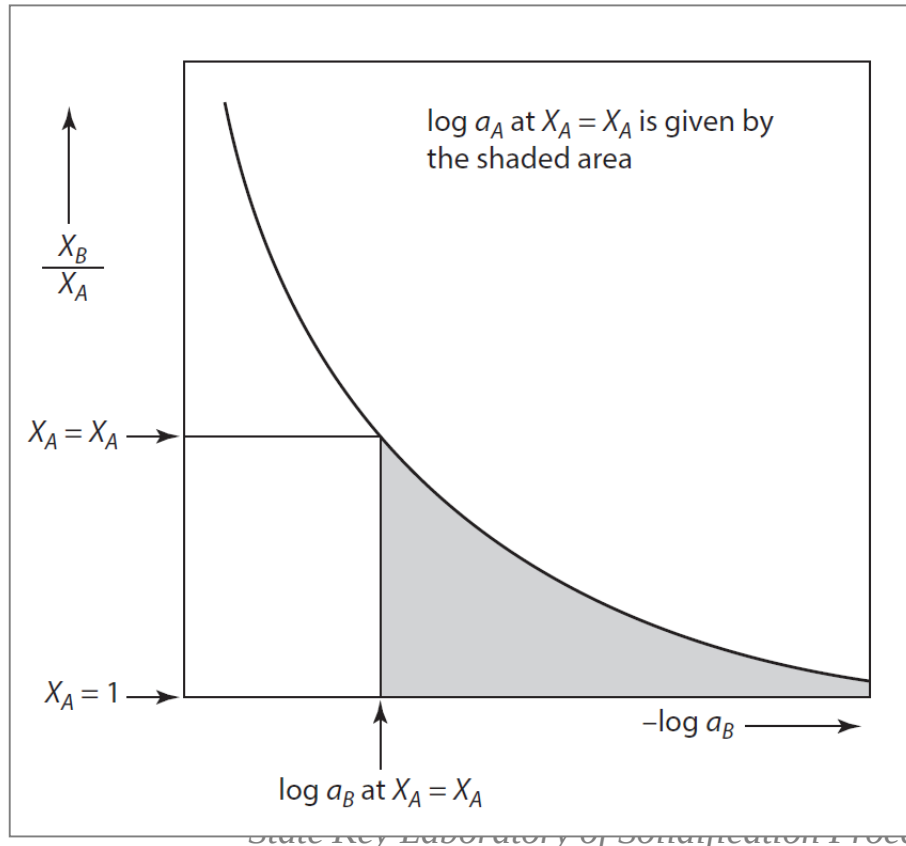
9.8. Application of the Gibbs-Duhem Relation to the Determination of Activity

1. $X_B \rightarrow 1$, ($X_A \rightarrow 0$), $a_B \rightarrow 1$, $\log a_B \rightarrow 0$, and $X_B/X_A \rightarrow \infty$.

Thus, the curve exhibits a tail to infinity as $X_B \rightarrow 1$.

2. $X_B \rightarrow 0$, ($X_A \rightarrow 1$), $a_B \rightarrow 0$, $\log a_B \rightarrow -\infty$, and $X_B/X_A \rightarrow 0$.

Thus, the curve exhibits a tail to minus infinity as $X_B \rightarrow 0$.



$\log a_B$ is the solid dark curve line

$\log a_A$ is the integration shaded area

Figure 9.10

A schematic representation of the variation of $\log a_B$ with X_B/X_A in a binary solution, illustrating the application of the Gibbs–Duhem equation to the calculation of the activity of component A.

9.8.1. The relationship between Henry's and Raoult's Laws

- ✓ Henry's law for the solute B in a binary A – B solution can be written as

$$a_B = \gamma_B X_B \quad \leftarrow \quad a_i = \frac{k^i X_i}{p_i^\circ} = k_{(i)} X_i$$

In terms of logarithms

$$\ln a_B = \ln \gamma_B + \ln X_B$$

Differentiation gives

$$d \ln a_B = d \ln X_B$$

9.8.1. The relationship between Henry's and Raoult's Laws

$$d \ln a_A = -\frac{X_B}{X_A} d \ln a_B$$

$$d \ln a_B = d \ln X_B$$



$$d \ln a_A = -\frac{X_B}{X_A} d \ln X_B = -\frac{X_B}{X_A} \frac{d \ln X_B}{X_B} = -\frac{dX_B}{X_A}$$

$$= \frac{dX_A}{X_A} = d \ln X_A$$

Integration gives



$$\ln a_A = \ln X_A + \ln(\text{constant})$$

or

$$a_A = \text{constant} \cdot X_A$$

9.8.1. The relationship between Henry's and Raoult's Laws

$$\ln a_A = \ln X_A + \ln(\text{constant}) \quad a_A = \text{constant} \cdot X_A$$

- ✓ But, by definition, $a_i = 1$ when $X_i = 1$, and thus, the integration constant equals unity. Consequently, in the range of composition over which the solute B obeys Henry's law, the solvent A obeys Raoult's law.
- ✓ As an example of this, see Figure 9.8, where Henry's law holds for the solute Ni up to about $X_{Ni} = 0.3$ and Raoult's law holds for the solvent Fe for $X_{Ni} < 0.3$.

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing

$$\Delta \bar{G}_A^M = \Delta G^M + X_B \frac{d\Delta G^M}{dX_A}$$

✓ Rearranging and dividing by X_B^2 gives

$$\frac{\Delta \bar{G}_A^M dX_A}{X_B^2} = \frac{X_B d\Delta G^M - \Delta G^M dX_B}{X_B^2} = d\left(\frac{\Delta G^M}{X_B}\right)$$



$$d\left(\frac{\Delta G^M}{X_B}\right) = \frac{\Delta \bar{G}_A^M}{X_B^2} dX_A$$

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing

- ✓ Integrating between $X_A = X_A$ and $X_A = 0$ gives

$$\Delta G^M = X_B \int_0^{X_A} \frac{\overline{\Delta G}_A^M}{X_B^2} dX_A$$

- ✓ Since $\overline{\Delta G}_A^M = RT \ln a_A$ the integral molar Gibbs free energy of mixing of A and B can be obtained directly from the variation of a_A with composition as

$$\Delta G^M = RTX_B \int_0^{X_A} \frac{\ln a_A}{X_B^2} dX_A$$

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing

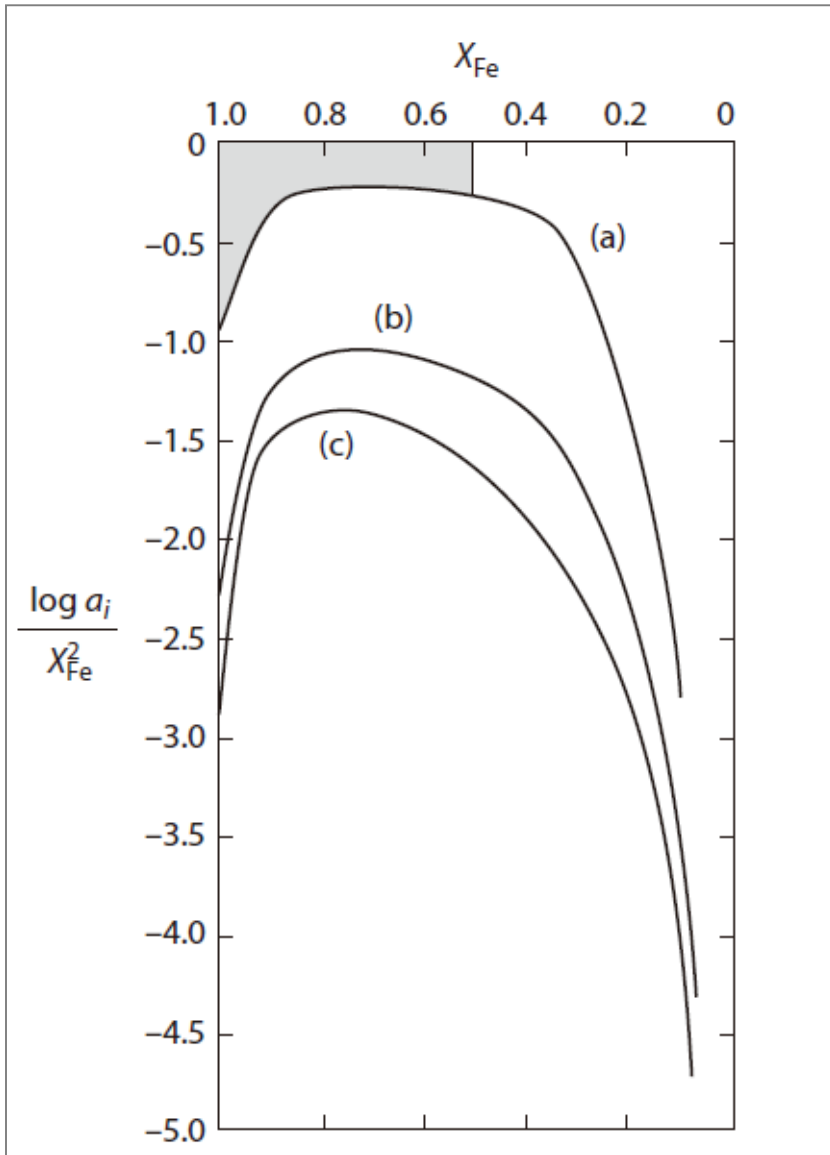
- ✓ The measured activities of Ni in Fe and Cu in Fe shown in Figures 9.8 and 9.9 can be used to obtain

$$\Delta G^M \text{ (in the system Fe-Ni)} = RTX_{Fe} \int_0^{X_{Ni}} \frac{\ln a_{Ni}}{X_{Fe}^2} dX_{Ni}$$

and

$$\Delta G^M \text{ (in the system Cu-Fe)} = RTX_{Fe} \int_0^{X_{Cu}} \frac{\ln a_{Cu}}{X_{Fe}^2} dX_{Cu}$$

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing



a: $\log a_{Cu}/X_{Fe}^2$

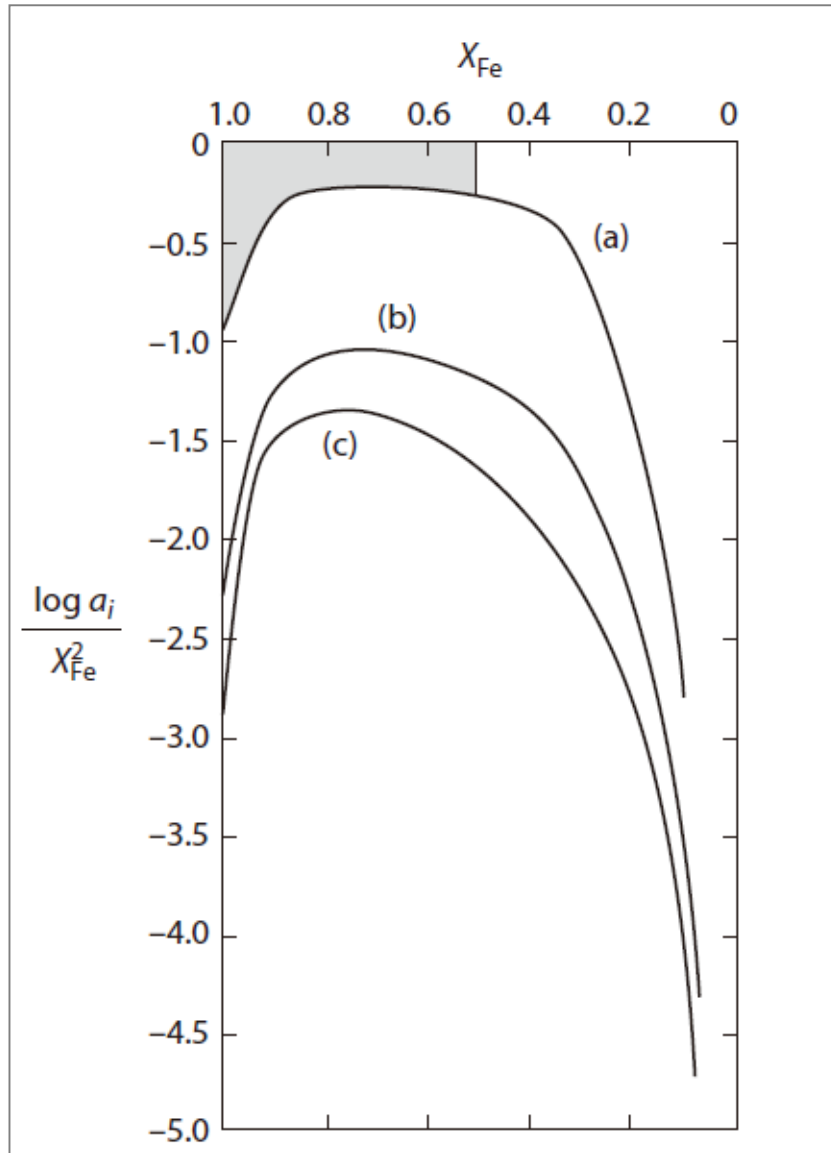
b: $\log X_i/(1-X_i)^2$

c: $\log a_{Ni}/X_{Fe}^2$

$$\text{Shaded area} = \frac{(\Delta G^M \text{ at } X_{Fe} = 0.5)}{2.303 \times 8.3144 \times 1823 \times 0.5}$$

Figure 9.11 Illustration of the direct calculation of the molar Gibbs free energies of mixing in the liquid systems iron– copper at 1550° C and iron– nickel at 1600° C.

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing



- ✓ Curve (a) is $(\ln a_{Cu})/X_{Fe}^2$ versus X_{Cu} and curve (c) is $(\ln a_{Ni})$ versus X_{Ni} .
- ✓ Curve (b) shows the variation of $(\ln X_i)/(1 - X_i)^2$ with X_i , which is the variation of the function for a component i which exhibits Raoultian behavior.

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing

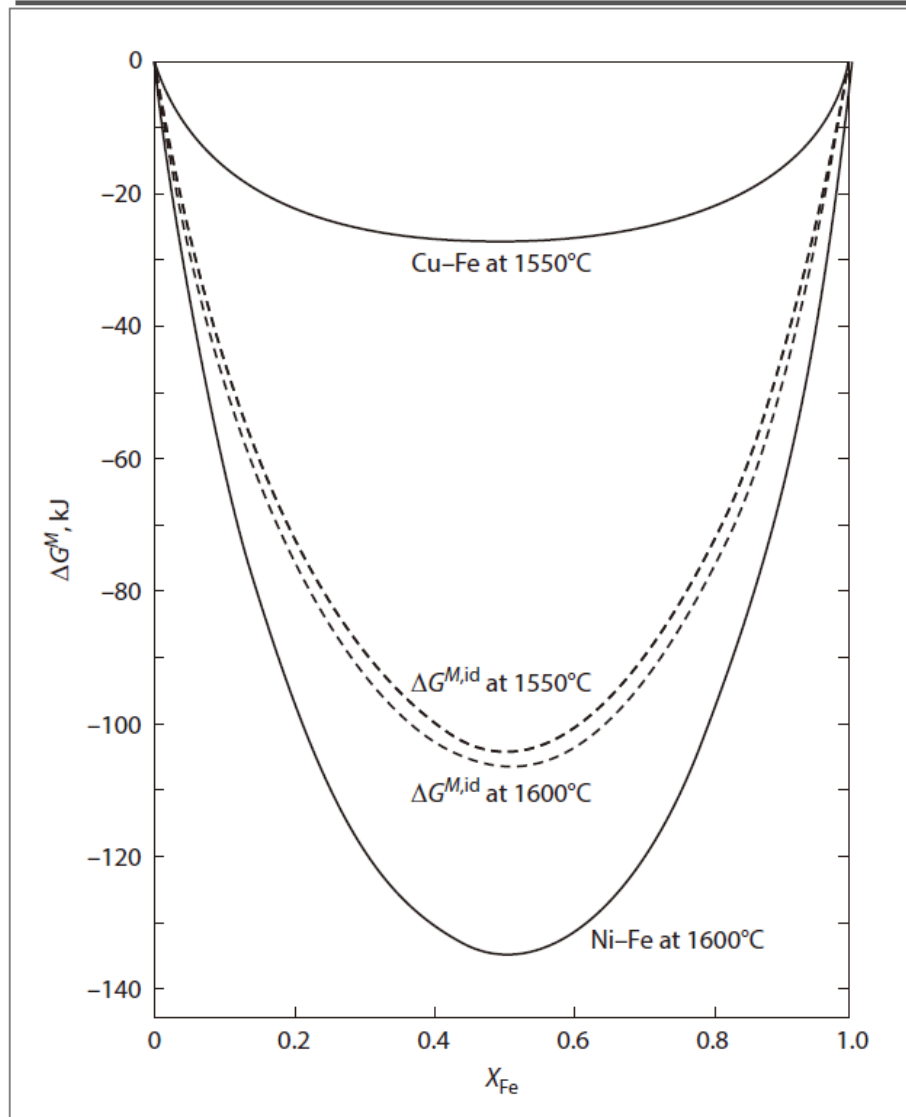


Figure 9.12 The molar Gibbs free energies of mixing in the liquid systems iron– copper at 1550° C and iron– nickel at 1600° C. Iron nickel displays negative deviations from ideality and iron copper displays positive deviations from ideality.

9.8.2. Direct Calculation of the Total Molar Gibbs Free energy of Mixing

- ✓ Here, the α function $\alpha_i = \ln \gamma_i / (1 - X_i)^2$ introduced by Darken and Gurry is applied to a solution which exhibits ideal ($\gamma_i = 1$) behavior (curve b); integration gives

$$\begin{aligned}\Delta G^M &= RT(1 - X_B) \int_0^{X_B} \frac{\ln X_B}{(1 - X_B)^2} dX_B \\ &= RT(1 - X_B) \left[\frac{X_B \ln X_B}{1 - X_B} + \ln(1 - X_B) \right] \\ &= RT \left[X_B \ln X_B + (1 - X_B) \ln(1 - X_B) \right] = RT \left[X_B \ln X_B + X_A \ln X_A \right]\end{aligned}$$

Homework

1. Use the Gibbs–Duhem equation to show that, if the activity coefficients of the components of a binary solution can be expressed as

$$\ln \gamma_B = \alpha_1 X_B + \frac{1}{2} \alpha_2 X_B^2 + \frac{1}{3} \alpha_3 X_B^3 + \dots$$

$$\ln \gamma_B = \beta_1 X_A + \frac{1}{2} \beta_2 X_A^2 + \frac{1}{3} \beta_3 X_A^3 + \dots$$

over the entire range of composition, then $\alpha_1 = \beta_1 = 0$, and that, if the variation can be represented by the quadratic terms alone, then $\alpha_2 = \beta_2$.

Homework

2. The activity coefficient of Zn in liquid Zn–Cd alloys at 435°C can be represented as

$$\ln \gamma_{Zn} = 0.875 X_{Cd}^2 - 0.30 X_{Cd}^3$$

Derive the corresponding expression for the dependence of $\ln \gamma_{Cd}$ on composition and calculate the activity of cadmium in the alloy of $X_{Cd} = 0.5$ at 435°C.

Homework

3. Tin obeys Henry's law in dilute liquid solutions of Sn and Cd, and the Henrian activity coefficient of Sn, γ_{Sn}° , varies with temperature as

$$\ln \gamma_{Sn}^{\circ} = -\frac{840}{T} + 1.58$$

Calculate the change in temperature when 1 mole of liquid Sn and 99 moles of liquid Cd are mixed in an adiabatic enclosure. The molar constant-pressure heat capacity of the alloy formed is 29.5 J/K

Thermodynamics of Materials & Phase transformations

The End

